

CITYTWIN · DESIGN PRECIS

CityTwin design precis

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ c1de765		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 update with adopted choices per CTW-DDR-001, numbers from CTW-CAL-001, parametric model and GA drawing CTW-DWG-001, TwinKit layout in the cabinet, pull paths for PotholeLog and DockHub
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, privacy rules, kiosk massing model, first-order numbers, safety, open questions)
0.1	2026-09-25	Amish Chadha	Initial scaffold

CityTwin design precis

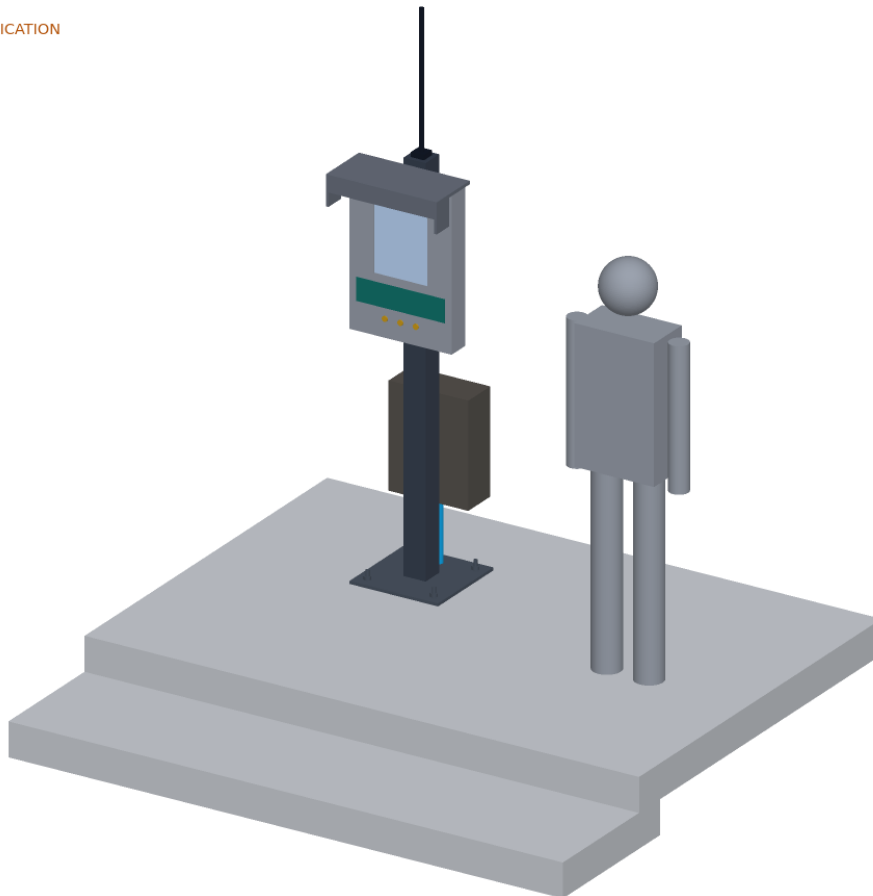
■ Summary

CityTwin is open software that runs on a TwinKit gateway and brings the lab's smart city nodes onto one map, plus a public street kiosk that shows the same data to residents. Nodes send counts and levels over LoRaWAN; CityTwin checks every record against a schema, stores it, publishes hourly open data files and draws the map. The kiosk is a steel post with a 13.3 in color e-paper screen at eye height, three push buttons, a notice plate that says what is measured and by whom, and a locked cabinet that holds the mains protection and the gateway.

The TRL 3 calculations (CTW-CAL-001) show that the data side has wide margin: the 46-node reference neighborhood sends 5,616 uplinks a day and loses 0.61 % to collisions and downlink blanking, and the store and open export are small. The kiosk takes 10.2 W from the mains with the gateway, weighs about 58 kg and stands up to a 35 m/s gust with a large margin. Four requirements are not met: a button press takes 19.8 s to show a new layer (R10); the kiosk parts cost \$739.00 against the \$600 budget, though within the \$750 recommended (R16); the TwinKit backup lasts only 1.63 h with an aged, cold pack (R18); and a street kiosk in sun or frost is outside the panel's 0 to 40 °C rating (R20), which is why the pilot is sited in shelter (R12, at risk). The general arrangement is drawing [CTW-DWG-001](#).

CityTwin

CONCEPT, NOT FOR FABRICATION



Grey: Person, 1.75 m (scale), sidewalk and curb for scale

Figure 1. CityTwin kiosk on a sidewalk, with a 1.75 m person for scale. Rendered from the parametric model `cad/src/model.py`.

■ How it works

1. **Collect.** The seven LoRaWAN node types (CurbCount, CrossSafe, LoadZone, HeatMap Node, FloodGauge, AirStreet, NoiseMap) send small uplinks to the TwinKit gateway's concentrator. PotholeLog uploads daily road summaries over depot Wi-Fi to the fleet operator's server, and DockHub syncs over LTE-M to its own back end. Under CTW-DDR-001 D11 the gateway pulls their published aggregates (road segment files once a day, dock-hour rows every hour) by outbound HTTPS, so it still accepts no inbound connection; the path is to be agreed with those projects. CrossSafe has no LoRaWAN uplink yet (its radio links the two sides of a crossing point to point), so its layer waits for that interface.
2. **Check.** A decoder for each node type turns the payload into named fields. Any field outside that type's schema is rejected and logged, so a changed or faulty node cannot push anything but counts and levels into the store.
3. **Store.** Records go into the TwinKit time-series database with node ID, location and time. Raw records are kept for two years and hourly aggregates indefinitely (D6); ten years of aggregates and two years of raw data take about 21 GB of the gateway's card.
4. **Apply the privacy rules.** Before anything is published, people and vehicle counts below 5 per hour (D6) become "fewer than 5", PotholeLog data is published only per road segment and per day, and DockHub data only per dock per hour. No card IDs, vehicle tracks or node-level raw streams leave the gateway.
5. **Publish.** Every hour the gateway writes CSV and GeoJSON files with a documented schema and pushes them one way to a public host under CC BY 4.0 (D7), about 0.86 MB of CSV a day. An OGC SensorThings API endpoint on the public host remains a suggestion for later work.
6. **Show.** The web map has one layer per node type, a time slider, node health and an "as of" timestamp. The kiosk controller pulls a pre-rendered 1600 x 1200 image of the selected layer from the gateway every 10 min, or when a button is pressed, and writes it to the e-paper. Each image carries its timestamp, so a stale screen during an outage says so.

CityTwin: LoRaWAN data flow, records per day (CTW-CAL-001 estimates, 46-node reference neighborhood)

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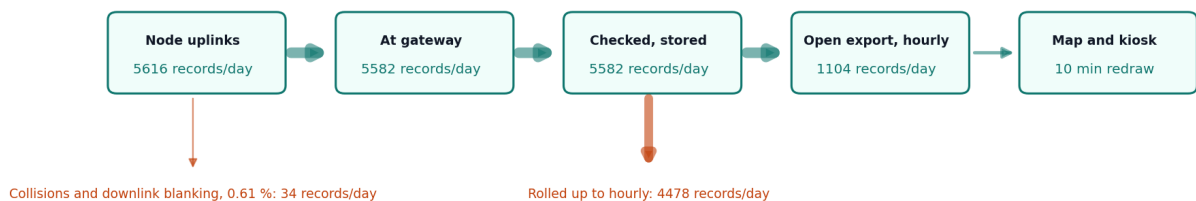


Figure 2. LoRaWAN data flow for the reference neighborhood, in records per day (CTW-CAL-001): 0.61 % lost to collisions and downlink blanking, and 5,582 stored records rolled up to 1,104 hourly records for the open export. PotholeLog and DockHub rows, pulled from their operators, are not shown.

■ Main components

Table 1. Main components. Numbers match `bom/bom.csv` and Figure 3.

#	Component	Choice	Notes
1	Base plate and anchors	400 x 400 x 12 mm galvanized plate, four M16 anchors on a 300 mm square	Into a footing or existing slab; 1.92 kN per anchor at the design gust
2	Post	100 x 100 x 4 mm square hollow section, 1.75 m	Carries head, hood, cabinet and antenna
3	Display head enclosure	Folded 2 mm aluminum, about 460 x 100 x 680 mm	IP54 target, vents underneath
4	Front window	6 mm UV-stabilized polycarbonate, anti-glare	Replaceable if scratched
5	E-paper display	13.3 in E Ink Spectra 6, 1600 x 1200 px, about 150 px per inch	Waveshare 13.3inch e-Paper HAT+ (E) : about 19 s full refresh, no partial refresh, 0 to 40 °C (manual)
6	Kiosk controller	Raspberry Pi Zero 2 W class, wired Ethernet to the gateway	Only pulls and draws images
7	Push buttons	Three 19 mm stainless buttons with LED rings, at about 1.12 m	Layer, time range, "about this sensor"
8	Data notice plate	400 x 100 mm, icons in the style of the open DTPR standard, QR code and short URL	Says what is measured, who is responsible and for how long data is kept
9	Sun and rain hood with light	Folded aluminum, 2 W LED strip with dusk sensor	Shades the screen by day, lights it at night
10	Services cabinet	Lockable steel, 360 x 160 x 460 mm, IP55, two 320 mm DIN rails, door on the back	Only place with mains voltage
11	Mains protection	6 A 30 mA RCBO and Type 2 surge protector	Installed or checked by an electrician
12	12 V DIN power supply	60 W, certified, SELV output	Feeds gateway and head
13	TwinKit gateway	TwinKit DIN gateway on the upper rail as in TWK-DWG-001 (294 mm of rail), LiFePO4 pack inside its UPS module	Costed in TwinKit (\$290.00); runs CityTwin software
14	LoRaWAN antenna	TwinKit antenna on a post-top bracket, tip at 2.38 m, coax to the gateway's SMA bulkhead	Low for a gateway; see open questions
15	Conduit and cabling	Mains conduit into the cabinet; 12 V and Ethernet up the post	
16 to 18	Software and templates	Map dashboard, open data export, governance templates	MIT and CC licensed; no parts cost

CityTwin: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Base plate and anchors
- 2 Post, 100 mm square steel
- 3 Display head enclosure
- 4 Front window, polycarbonate
- 5 E-paper display, 13.3 in color
- 6 Kiosk controller
- 7 Push buttons (3)
- 8 Data notice plate
- 9 Sun and rain hood with light
- 10 Services cabinet, lockable
- 11 Mains protection (RCBO, surge)
- 12 12 V DIN power supply
- 13 TwinKit gateway with backup
- 14 LoRaWAN antenna (TwinKit)
- 15 Conduit and cabling

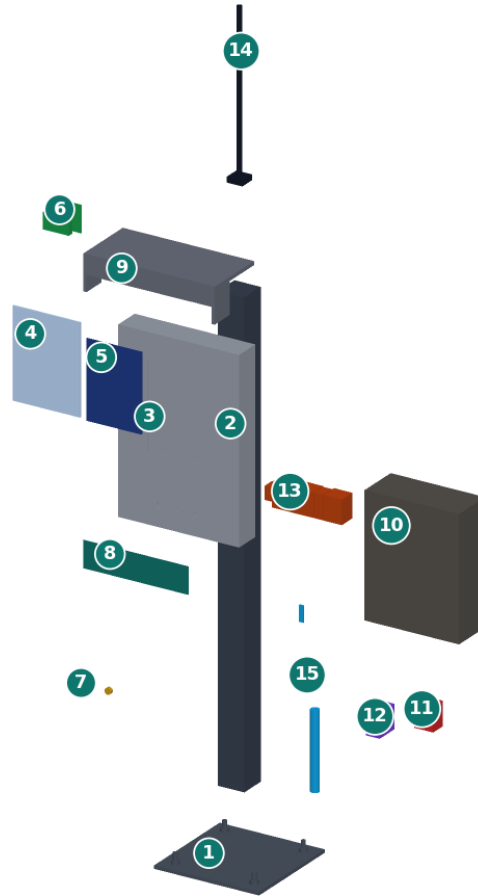
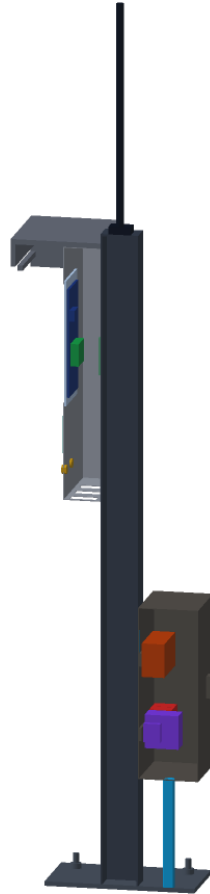


Figure 3. Exploded view with BOM numbers.

CityTwin: cutaway (section on the kiosk centerline, looking from the right)

CONCEPT, NOT FOR FABRICATION



Head: window (4), e-paper (5), controller (6). Cabinet: gateway (13), supply (12), protection (11).

Figure 4. Section on the kiosk centerline, looking from the right: window (4), e-paper (5) and controller (6) in the head; TwinKit gateway (13) on the upper rail, power supply (12) and mains protection (11) on the lower rail of the cabinet.

■ Key numbers

All values are from the TRL 3 calculation note CTW-CAL-001, which states every assumption; the tags refer to lines of its script.

Table 2. Data load for the reference neighborhood (CTW-DDR-001 D9).

Node type	Nodes	Uplinks per node per day	Uplinks per day	Basis
CurbCount	8	96	768	14 B per 15 min bin (CBC-CAL-001)
CrossSafe	4	24	96	Hourly summary (CityTwin assumption; no LoRaWAN uplink defined in CrossSafe)
LoadZone	12	124	1,488	100 state changes and 24 heartbeats, 12 B (LDZ-CAL-001)
HeatMap Node	6	96	576	20 B per 15 min (HMN-CAL-001)
FloodGauge	4	96	384	15 min normal mode; 1 min in events (FLG-CAL-001)
AirStreet	6	288	1,728	20 B per 5 min (AST-CAL-001)
NoiseMap	6	96	576	22 B per 15 min (NSM-CAL-001)
Total	46		5,616	R2 met on paper

Table 3. System and kiosk numbers (CTW-CAL-001).

Quantity	Value	Requirement
Radio airtime at SF9	1,309 s a day, 0.189 % of each of 8 channels	
Radio loss	0.38 % collisions plus 0.23 % while the gateway sends LoadZone sign downlinks; 0.94 % plus 0.23 % in a storm peak hour	R2 met on paper
Gateway capacity	14,075 records a day of 20 B at SF9 before loss reaches 1 %	R2
Storage	2.89 GB a year raw; 20.9 GB for two years raw and ten years of hourly aggregates	Fits 56 GB with TwinKit's own data
Open export	7,128 rows a day, 0.86 MB CSV and 2.85 MB GeoJSON	R6
Worst data age	25.9 min on the kiosk, 15.6 min on the web map	R8 met on paper, thin margin
Button to new layer	19.8 s (0.79 s transfer plus 19 s refresh)	R10 not met
Text and night light	7 mm capitals are 41 px and 16.0 arcmin at 1.5 m; about 1,459 lx from the 2 W hood light	R9 met on paper
Hood shading	13 %, 31 % and 63 % of the window shaded with the sun at 30°, 45° and 60° in front	
Head and panel, pilot	Panel 40.1 °C at 40 °C air in shade	R12 at risk
Head and panel, street	About 79 °C in low sun at 45 °C air; 13.5 W to keep an insulated panel bay at 0 °C in -20 °C air	R20 not met
Cabinet	8.11 W inside; air 4.4 K above ambient in shade, 18.8 K in sun; pack 0.6 K under its 45 °C charge limit at 40 °C air	R12 at risk
Power	10.2 W from the mains, 90 kWh a year; 12 V peak 26.7 W on a 60 W supply	R15 met on paper
Wind at 35 m/s	792 N, 826 N·m at the base; 26.2 MPa in the post with a 1.5 factor; 1.92 kN per anchor	R13 met on paper
Mass	58.2 kg	Two-person lift
Backup	2.40 h new, 1.63 h aged at 0 °C (TwinKit)	R18 not met in the worst case
Kiosk parts cost	\$739.00	R16 not met against \$600; within the \$750 recommended
Kiosk and gateway	\$1,029.00	R17 reported

■ Key design choices

Items D1 to D10 in CTW-DDR-001 are adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. D11 is a new proposal from this session.

- **Color e-paper rather than an LCD (D2).** E-paper is readable in direct sun, draws almost nothing between refreshes and keeps its image in an outage. A 1,000-nit outdoor LCD would respond instantly and allow touch, but costs more, draws tens of watts and needs cooling. A monochrome panel would refresh faster but lose the color layers. Adopted: color e-paper for the street kiosk, with an indoor LCD variant for libraries and city hall lobbies (not modeled at TRL 3). The cost is R10, not met.
- **Sheltered pilot (D3).** The panel is rated 0 to 40 °C. The pilot kiosk stands where the screen gets no direct sun; the heater and shading study for a street kiosk is CTW-CAL-001 section F.
- **Buttons, not a touch screen (D8).** Three stainless buttons survive vandalism and weather and suit e-paper's slow refresh; the QR code hands richer use to the visitor's phone.
- **Gateway inside the kiosk cabinet (D4).** One site, one mains feed and one locked box for a pilot. The TwinKit gateway sits on its own DIN rail exactly as TwinKit lays it out, with a coax lead to the post-top antenna in place of its whip. The cost is a low antenna (tip at 2.38 m), which shortens LoRaWAN range; a rooftop gateway is the choice for a full neighborhood.
- **Mains power (D5).** Kiosk and gateway need 0.246 kWh a day. A solar head-only variant is kept for sites without power (not modeled at TRL 3).
- **One-way publishing and outbound pulls (D10, D11).** The gateway pushes open data files out and fetches sibling aggregates out, and accepts no inbound connections from the internet, in line with TwinKit's advice to keep the gateway off public networks until it is hardened.
- **Privacy rules in the gateway, not only in the nodes (D10).** Nodes already send counts and levels only; enforcing a schema and small-count suppression (5 per hour, D6) again at the gateway protects against a changed node or a future node type that is less careful.
- **Open data license (D7).** CC BY 4.0.
- **Budget (D1).** `budget_usd` (\$600) covers the kiosk; the gateway is costed in TwinKit. The recommended new figure of \$750 awaits Amish.

■ Safety

Safety: The services cabinet contains mains voltage (230 V or 120 V). The mains feed, RCBO, surge protector and earthing must be installed or checked by a qualified electrician and follow local electrical code. Keep the cabinet locked, fit an earth bond to the post, head and hood, and carry only 12 V SELV up to the head.

Safety: The TwinKit gateway contains a LiFePO4 backup pack. Use a pack with a built-in BMS and fuse, charge it only through the TwinKit UPS module, and keep it within its temperature limits. CTW-CAL-001 finds the sealed cabinet 18.8 K above the air in full sun: at 45 °C air the pack would be near 64 °C, and in frost it cannot charge. The UPS charger's 0 to 45 °C lockout must never be defeated; the pilot site must keep the cabinet out of direct sun.

Safety: The kiosk is a roughly 58 kg steel post beside a walkway. Install it only with the asset owner's permission and a permit, on a footing designed for local wind and impact loads, with no sharp edges or protruding corners at head height, and keep a clear walkway width around it as local rules require. Lift it with two people or a hoist.

Safety: Privacy by design: no images, audio recordings or personal identifiers are taken in, stored or published; only aggregate counts or levels. Check local data protection law, and carry out a privacy impact assessment before any deployment. CityTwin is not a public warning system: do not rely on it for flood or emergency alerts.

■ Open questions

- [] Pull paths for PotholeLog segment files and DockHub hourly aggregates (D11): agree formats and hosts with those projects.
- [] A LoRaWAN uplink from CrossSafe, whose radio now runs point to point between the two sides (R1).
- [] Button response (R10): accept about 20 s with an LED acknowledgment and restate R10, show a small "loading" cue first, or use an LCD in the indoor variant. Proposed, awaiting Amish.
- [] Backup time (R18): restate R18 at 1.5 h, or ask TwinKit for a larger pack. Proposed, awaiting Amish.
- [] Street climate (R20): a heated, insulated panel bay (13.5 W at -20 °C), a north-facing or shaded face, a sun shield and vents for the cabinet, or a wider-range panel.
- [] Antenna height needed for the reference neighborhood with the gateway in the cabinet.
- [] Which languages and scripts the kiosk shows, and how residents ask questions or report a fault.
- [] First partner city or neighborhood, and co-design sessions with residents on what the kiosk should show (CTW-DDR-001 O1).

Concept media: [blueprint sheet](#), [interactive 3D model](#). General arrangement: [CTW-DWG-001](#). Calculations: [CTW-CAL-001](#).